Automatic GFL/GFM Mode Switching

A unified PF–SSSA–TS framework with screened reference-mode transitions

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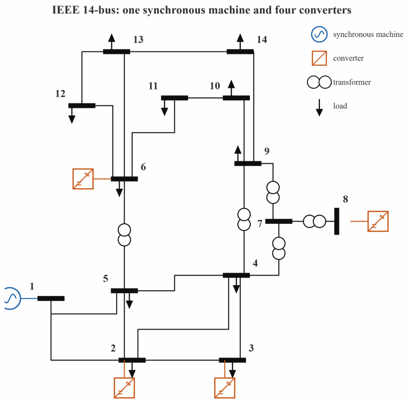


IEEE 14-bus • 1 synchronous machine • 4 dual-mode converters

Problem, research question, and contributions

The operating problem

GFL converters need an external angle reference; GFM converters can create one, but reserving every converter for GFM duty is costly and can introduce adverse mode interaction. When the last synchronous reference trips, the operator must decide more than simply “GFL or GFM.”

How many should form, which ones, and by what transition route?

Two gaps matter: linear admissibility does not prove that the present state can reach a configuration, and benefit is not monotone in the number of GFM units.

What this project contributes

1. One unified mathematical path from AC power flow through SSSA to switched time-domain simulation.
2. A dual-mode coordinate superset with active-state projections and a current-continuity-gated GFL↔GFM assignment.
3. Exhaustive screening of all $2^4 - 1 = 15$ non-empty GFM sets at their own equilibria, followed by transition certification.
4. A five-policy falsification study: zero GFM is infeasible, four pinned GFM units hunt, and staged automatic switching completes 250 s.

Motivation: Cui et al., *IEEE TPS*, 2025; Ding et al., *IEEE TSTE*, 2025.

One mathematical stack: PF → SSSA → TS

Network equilibrium and bus semantics

$$F_{\mathrm{PF}}(z) = 0, \qquad J_{\mathrm{PF}}\Delta z = -F_{\mathrm{PF}},$$
$$g(x, V) = Y_{\mathrm{net}} V - I_{\mathrm{bus}}(x, V) = 0.$$

REF fixes $|V|$, θ; PV fixes P, $|V|$; PQ fixes P, Q. IEEE-14 converges in 5 Newton steps with $\|F_{\mathrm{PF}}\|_\infty = 6.34 \times 10^{-15}$ pu.

Exact DAE linearisation

$$\begin{bmatrix}\Delta\dot{x}\\ 0\end{bmatrix} = \begin{bmatrix} f_x & f_y \\ g_x & g_y \end{bmatrix}\begin{bmatrix}\Delta x \\ \Delta y\end{bmatrix}, \qquad A = f_x - f_y(g_y \backslash g_x).$$

One network-angle gauge coordinate is removed. Paired left/right eigenvectors give participation; $\zeta = -\sigma/\sqrt{\sigma^2 + \omega^2}$ is compared with the declared 0.05 floor.

One coupled implicit time step

$$R_x = x_{n+1} - x_n - \frac{h}{2}[f(x_n, y_n) + f(x_{n+1}, y_{n+1})] = 0,$$
$$R_y = g(x_{n+1}, y_{n+1}) = 0.$$

Damped Newton solves both residuals together. Adaptive step doubling estimates local error; accepted steps land exactly on scheduled events. Domain-invalid trials are rejected and unresolved transitions fail closed.

A switch is an assignment, not a time step

Inactive controller coordinates are re-initialised at the current terminal operating point. Commit requires $\|I^+ - I^-\|_\infty \le 10^{-10}$ pu and a right-limit KCL solve.

Defensible decision parameters

Normalised severity and hysteresis

$$J_V = \frac{\big||V_i| - V_i^\star\big|}{0.10\,\mathrm{pu}}, \qquad J_f = \frac{|f_{\mathrm{COI}} - 60|}{0.50\,\mathrm{Hz}},$$

$$S = \mathrm{sat}_{[0,1]}\left(\tfrac{1}{2}J_V + \tfrac{1}{2}J_f\right).$$

Equal weights combine local voltage and system frequency after normalisation. The hysteresis and dwell conditions are

$$S \geq 0.65 \text{ for } 0.10\,\mathrm{s} \qquad (\mathrm{GFL} \to \mathrm{GFM}),$$
$$S < 0.35 \text{ for } 1.00\,\mathrm{s} \qquad (\mathrm{GFM} \to \mathrm{GFL}).$$

The gap prevents chatter; the unequal dwell makes support fast and release slow.

Prediction frozen before measurement

The non-ideal DC source is specified as a second-order, maximally-flat response:

$$\zeta^\star = \frac{1}{\sqrt{2}}, \qquad \tau_s = 5.00\,\mathrm{ms}.$$

This predicts, before the assembled spectrum is evaluated,

$$f \approx 15.5\,\mathrm{Hz}, \qquad \zeta = 0.7071.$$

The measured four-converter result is

$$f = 15.18\text{–}15.66\,\mathrm{Hz}, \qquad \zeta = 0.7072\text{–}0.7079.$$

The agreement is therefore a prediction check, not parameter fitting.

Interpretation rule

Parameter consequences are checked after selection; no parameter, event or acceptance threshold is changed in response to the observed curves.

Mathematical bases: IEEE Std 1110; Kundur (1994); Kodsi–Cañizares (2003); MATPOWER IEEE-14 data; stated project derivations.

Independent validation I: power flow and small signal

Identical mapped inputs: proposed framework vs PSAT 2.1.11 GATED

quantity	pair	IEEE 14	RTS-24
network contract max $\lvert\Delta Y\rvert$	Ours−PSAT	3.97×10^{-15}	2.84×10^{-14}
power flow max $\lvert\Delta V\rvert$ [pu]	PSAT−Ours	6.661338×10^{-16}	4.441×10^{-16}
power flow max $\lvert\Delta \theta\rvert$ [deg]	PSAT−Ours	3.552714×10^{-14}	6.750×10^{-14}

Tolerances declared *before* the comparison: $\Delta V < 10^{-6}$ pu, $\Delta\theta < 10^{-4}$ deg. The required gate passes with ten orders of magnitude to spare — the two solvers agree at machine precision, not merely within a tolerance.

Small signal, five-machine classical IEEE 14 GATED

Four oscillatory modes give $|\Delta\lambda| = 1.000 \times 10^{-6}\ \mathrm{s}^{-1}$ on *every* mode. This uniform offset is the documented regularisation of the reference analysis; the physical frequencies match to the printed precision: 1.83350, 1.58440, 1.53137, 1.35374 Hz on both sides.

Validation II: transient trajectories

Proposed framework vs PSAT 2.1.11 (IEEE 14-bus) GATED

Metric	Max Difference	Tolerance Gate
$\Delta\delta_{\mathrm{COI}}$ [deg]	9.571×10^{-3}	< 0.05
$\Delta\omega$ [pu]	3.816×10^{-6}	$< 1.0 \times 10^{-4}$
ΔP_e [MW]	4.217×10^{-2}	< 0.10
$\Delta\lvert V\rvert$ (fault bus) [pu]	3.179×10^{-5}	$< 1.0 \times 10^{-3}$

Pre-declared tolerances on dynamic state trajectories are satisfied across the full 15 s simulation horizon with substantial margin.

Trajectory Agreement Insights

- **Close trajectory agreement**: Generator rotor angles δ_{COI}, speeds ω, active power outputs P_e, and bus voltages $|V|$ closely trace PSAT reference trajectories throughout fault and post-fault periods.
- **Acceptance evidence**: The complete comparison horizon contains zero unconverged steps and every measured difference remains inside its gate.
- **Validated baseline**: Establishes an independent reference on classical machines before incorporating advanced dual-mode IBR converters.

Every pre-declared PSAT trajectory gate is satisfied before the analysis is extended to dual-mode IBR converters.

Same network, four converters, one switching contract

The PSAT validation ran classical machines at buses 1, 2, 3, 6, 8. This study keeps the machine at bus 1 and replaces buses 2, 3, 6, 8 by four dual-mode converters on the same network and operating base.

One coordinate layout, two regimes — exclusive

grid-following	10 active	7 controller states frozen
grid-forming	11 active	6 controller states frozen
tripped	0 active	everything held

17 coordinates per converter; composite $6 + 4 \times 17 = 74$ coordinates. The 17-vector is a *superset*, not a simultaneously active state: only the selected controller branch evolves in each regime.

The transfer itself

A mode change is an *assignment*, not a trapezoidal step, and it commits only if terminal current is continuous to 10^{-10}. The parameterisation may jump; the current the network sees cannot.

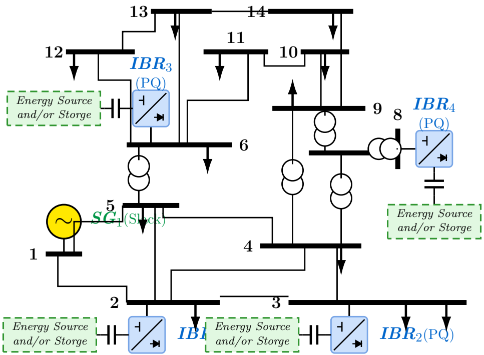
The severity index that triggers it

$$S = \mathrm{sat}_{[0,1]}\left(\tfrac{1}{2}J_V + \tfrac{1}{2}J_f\right)$$

$$J_V = \frac{\big||V_i| - V_i^\star\big|}{0.10\,\mathrm{p.u.}}, \qquad J_f = \frac{|f_{\mathrm{COI}} - 60|}{0.50\,\mathrm{Hz}}$$

$V_i^\star$ is the healthy power-flow voltage; the equal weighting is a project choice frozen before any result was read. Commit at $S \geq 0.65$ held 0.10 s; release at $S < 0.35$ held 1.00 s — quick to add support, deliberately slow to release it. Four further sub-indices are computed but kept diagnostic: they enter no gate and form no aggregate by design.

Study network



The SG holds the healthy-system angle reference. After its trip, an authenticated GFM converter must own the island reference; all devices remain coupled through $g(x, V) = Y_{net}V - I_{bus}(x, V) = 0$.

The spectrum, at the two operating points that matter

Machine online, all converters following

44 physical coordinates of 45 (one angle removed by the network-angle quotient). The electromechanical mode is the one the case contract speaks about:

λ	$-1.146045 \pm 7.254946\mathrm{j}$
f	1.154661 Hz
ζ	0.156033
dominant state	machine speed

against the declared floor $\zeta_{\min} = 0.05$ — a factor of three of margin. Slowest root -0.174920 (machine field flux).

Machine tripped, one converter forming

40 physical of 41; 30 printed rows, conjugate pairs once; *entirely* in the left half plane — the island the converters hold is small-signal stable.

	λ	dominant
fastest	-2075.371442	inner current
DC family	$-98.41 \pm 98.38\mathrm{j}$	DC link
	. . . 3 more pairs	DC link
PLL pair	$-0.588912 \pm 2.171749\mathrm{j}$	PLL angle
slowest	$-0.328014 \pm 0.448860\mathrm{j}$	voltage loop

The slowest root *is* the decay-rate margin the selector ranks on.

Two limits stated, not buried OPEN

9 of the 30 rows have their two leading participations within 0.02, so the named state there is *not* a sole attribution — every table carries that mark per row. And this is one linearisation point: it is not, and must not be read as, a stability certificate for the switched trajectory.

Predicted first, then measured: the DC link

What was predicted, before the run

The converter's DC side was an ideal source. Replacing it by a real source behind a resistance and an inductance adds one state per converter, and its time constant is not free: requiring the flattest possible response fixes the damping at $\zeta^\star = 1/\sqrt{2}$ and therefore the time constant at 5.00 ms. From that alone the pole pair was predicted, on paper, at

$$-97.7 \pm 97.7\,\mathrm{s}^{-1}, \quad f \approx 15.5\,\mathrm{Hz}, \quad \zeta = 0.7071.$$

Two feasibility bounds were declared at the same time, and the realised margin against the binding one is 2.07–2.15. The source voltage is not chosen either — the dispatch forces it, giving 1.0269–1.0448 p.u. across the four converters.

What the assembled system then returned GATED

	predicted	measured
f [Hz]	≈ 15.5	15.18–15.66
ζ	0.7071	0.7072–0.7079

Four converters, four pairs, every one of them where the derivation said it would be. Participation splits exactly 0.500/0.500 between the two DC coordinates and is 0.000 everywhere else, so each pair *is* one converter's DC circuit and nothing else.

And the coupling was measured, not assumed OPEN

The AC equations read the DC states with sensitivity *exactly zero*; the DC equations do read the AC. The coupling is one-way — which also means a sagging DC link never limits an AC command in this model, so nothing here covers DC-limited ride-through.

How many converters, and which — computed

Candidate Set Admissibility

size	grid-forming set	margin [s^{-1}]	outcome
1	bus 8	−0.155437	admissible
1	bus 6	−0.252025	admissible
1	bus 3	−0.316711	admissible
1	bus 2	−0.328014	admissible
2	buses 2 + 6	−0.561723	admissible
2	**buses 3 + 6**	**−0.568138**	**best**
2	four other pairs	–	limit hit
3	all four triples	–	no equilibrium
4	all converters	−0.482909	admissible

What the enumeration proves

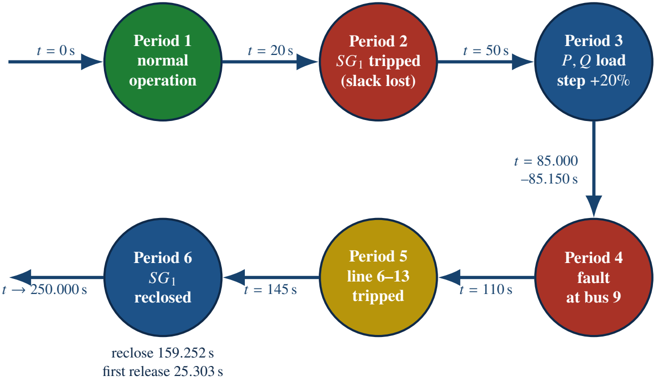
15 candidate sets evaluated, 7 admissible — 4 of 4 at size 1, 2 of 6 at size 2, 0 of 4 at size 3, 1 of 1 at size 4.

- **Non-monotonic**: 1-GFM sets pass, 3-GFM fail, optimum sits at 2-GFM ($-0.5681\,s^{-1}$).
- **Optimal pairing**: Buses 3 + 6 outperform single or 4-converter configurations.
- **Analytical proof**: Sizing cannot follow heuristic rules of thumb; must be rigorously computed.

Necessary, not sufficient OPEN

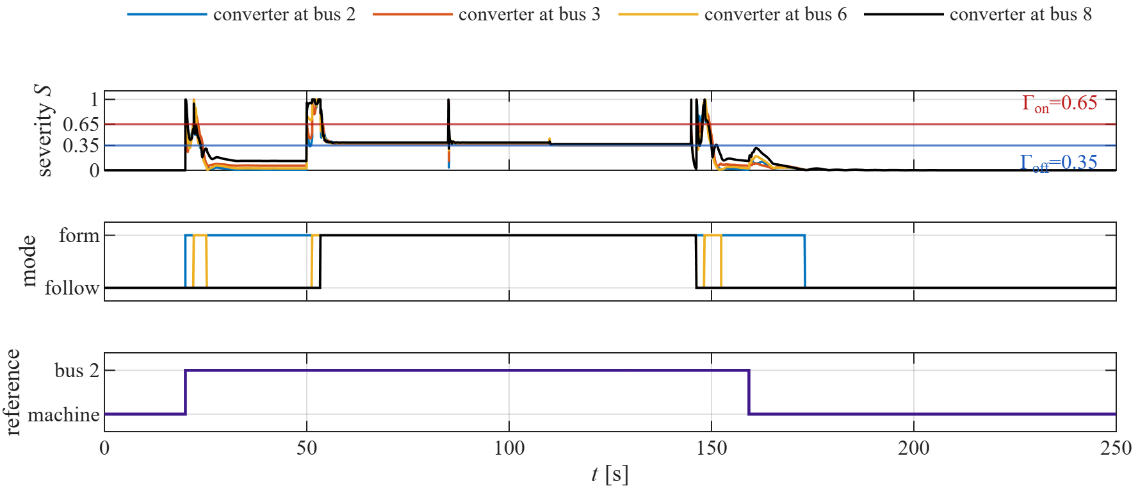
The 4-converter set passes the linear SSSA screen yet *fails* dynamically: pinned from the trip it terminates at $t = 25.488\,s$ due to unmitigated hunting.

The disturbance chronology: six periods



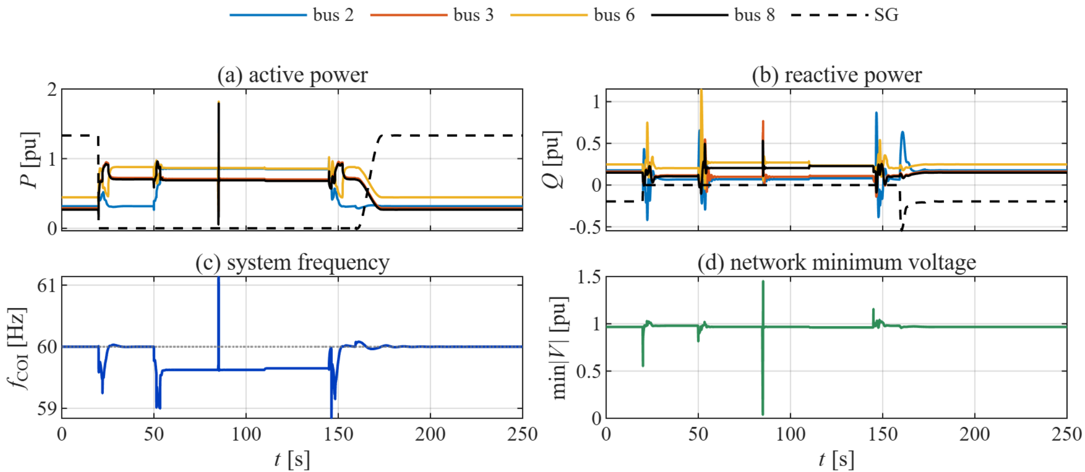
Green: undisturbed. Red: a contingency that removes a source or faults a bus. Yellow: a topology change. Blue: a restorative action. The six scheduled instants are case-defined; the reclose and first release instants are *read from the accepted event log* of the run — they are results, not inputs.

The supervisor's decision, over the whole run



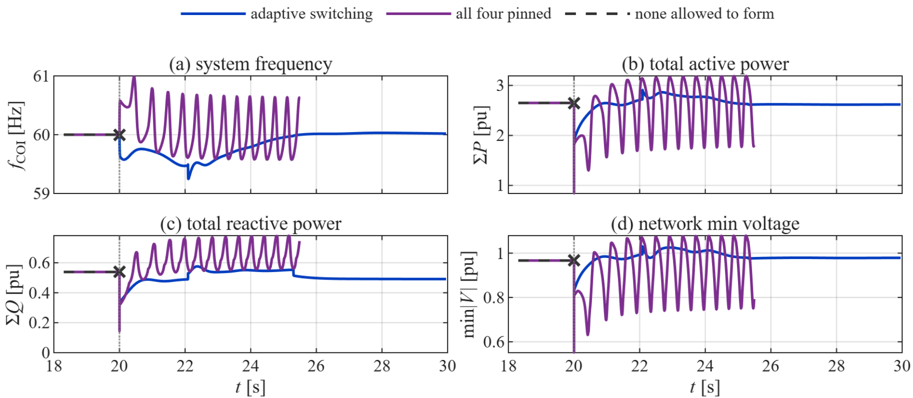
The severity index crosses the commit threshold *only* at the disturbances, and the asymmetric dwell prevents chatter. Peak commitment is 4 converters; the run ends with *none* grid-forming and the machine holding the reference again.

Electrical response on the same time axis



The machine (dashed) drops to zero at the trip and returns at the reclose; the converters take up the difference. Frequency shows both under- and overshoot at the disturbances, then recovers to 60.000001 Hz. Voltage collapses only during the bolted fault and returns to 0.9667–1.0575 p.u..

Switching is necessary, not merely better



None allowed to form — with zero voltage-forming capability, the post-trip solve is *refused* at $t = 20$ s (43 samples). **All four pinned** — rigid GFM mode interaction causes rapid power and frequency hunting, terminating at $t = 25.488$ s. **Adaptive switching** — rides through all six scheduled disturbance events.

Multi-signal dynamic evidence:

- Frequency f_{COI} settles cleanly at nominal 60 Hz without hunting.
- Total power $(\Sigma P, \Sigma Q)$ shares the islanded load while min $|V|$ recovers after each cleared contingency.
- Settled deviation improved by 3.28× vs static GFM during fault.

Established here — and openly not claimed

Established GATED

- power flow and time-domain results agree with an independent third-party tool on two networks, under tolerances declared in advance — power flow at machine precision. The same gate *refuses* a second third-party tool, and that refusal is printed
- the composite Jacobian's block structure follows from a proposition with an *iff* converse, not from an assumed sparsity pattern
- on this resource mix at least one grid-forming converter is *necessary* for a post-trip trajectory to exist at all
- the admissible grid-forming set is non-monotone in its size, with an interior optimum — so it must be computed, not assumed
- the 250 s chronology closes: the machine recloses on merit, every converter is handed back, and the terminal frequency lands within 10^{-6} Hz of nominal

Not claimed OPEN

- **no operational readiness** — convergence is not grid-code compliance, and the extrema are model excursions, not ratings
- **no selector-against-selector** claim: this is switching against *not* switching, on this resource mix
- **no multi-machine** result — that path fails closed at four points and no numbers are assumed in their place
- four of the six severity sub-indices are kept diagnostic: they enter no gate and form no aggregate by design
- the DC mode sits at the resolution limit of a phasor method; a firmer link belongs to an electromagnetic-transient study
- unresolved model limitations are disclosed explicitly and none is used to support an operational-readiness claim